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# The p38 MAP kinase pathway as a therapeutic target in inflammatory disease

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## Abstract

The p38 MAPK signalling pathway plays an important role in inflammation and other physiological processes. Specific inhibitors of p38  $\alpha$  and  $\beta$  MAPK block production of the major inflammatory cytokines (i.e. tumour necrosis factor- $\alpha$  and interleukin-1) and other proteins (e.g. cyclooxygenase-2), and are anti-inflammatory in animal models of disease. A major function of the pathway is post-transcriptional control of inflammatory gene expression. Many of the mRNAs are unstable (or untranslatable) because of AU-rich elements in the 3'untranslated region. Signalling in the p38 pathway counteracts these and stabilizes the mRNAs by preventing their otherwise rapid de-adenylation.

## Abbreviations

**ARE**, AU-rich element; **ATF**, activating transcription factor; **COX**, cyclooxygenase; **ERK**, extracellular signal-regulated kinase; **IL**, interleukin; **JNK**, c-jun N-terminal kinase; **LPS**, lipopolysaccharide; **MAPK**, mitogen-activated protein kinase; **MAPKAPK**, MAPK-activated protein kinase; **MKK**, MAP kinase kinase; **MNK**, MAPK signal integrating protein kinase; **MSK**, MAPK and stress kinase-activated protein kinase; **NFκB**, nuclear factor-κB; **TNF**, tumour necrosis factor; **TTP**, tristetraprolin

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## Introduction

Early in the discovery of the p38 signalling pathway, specific inhibitors of the p38 mitogen-activated protein kinase (MAPK) were identified. These pyridinyl imidazole compounds were found empirically to inhibit production of tumor necrosis factor (TNF) $\alpha$  and interleukin (IL)-1 by lipopolysaccharide (LPS)-stimulated cells. They were subsequently shown to bind a protein that proved to be p38 $\alpha$  MAPK [1]. The most widely used of these agents are SB203580 and SB202190 (Figure 1). Because inhibiting p38 MAPK suppresses production of key mediators, it was an obvious target for therapy of chronic inflammatory disease. Inhibitors are being developed with a view to treating common diseases such as rheumatoid arthritis, Crohn's disease, chronic obstructive pulmonary disease and psoriasis.

Inhibiting the MAPK pathway as an anti-inflammatory therapy might not necessarily be best achieved by inhibiting p38 MAPK itself; targets upstream or downstream could also be blocked. As discussed below, the pathway is involved in processes other than inflammation. If inhibition is to be effective and safe, it must not compromise normal physiology, and the pathological inflammation should be highly dependent upon the pathway. Interference with other processes might be avoided by intervening downstream where the pathway impinges on inflammatory gene expression.

There are two issues to discuss: firstly, what inflammatory mechanisms does the p38 MAPK pathway strongly regulate; and secondly, what other processes does it control?

MAPKs are proline-directed serine and threonine protein kinases that are activated by dual

specificity kinases by phosphorylation of threonine and tyrosine in a Thr-Xaa-Tyr motif in a loop near the active site (Figure 2) [2,3\*]. The three types are defined by the intervening amino acid. In the extracellular signal-regulated kinases (ERKs), the amino acid is Glu; in c-jun N-terminal kinases (JNKs), it is Pro; and in the p38 MAPKs, it is Gly.

There are four p38 MAPKs:  $\alpha$  and  $\beta$ , which are 75% homologous, and  $\gamma$  and  $\delta$ , which are more distant relatives [2]. All are activated by the same upstream kinases — MAP kinase kinase (MKK)3 or MKK6 — in response to inflammatory and stressful stimuli. The pyridinyl imidazole inhibitors SB203580 and SB202190 only antagonise  $\alpha$  and  $\beta$  forms and compete for the ATP-binding pocket. Thr106 is important for the specificity of this interaction [3\*]. A major substrate of p38 $\alpha$  and  $\beta$  is MAPK-activated protein kinase (MAPKAPK)-2 (Figure 2) [2]. The p38 $\gamma$  and  $\delta$  forms are less effective at activating the downstream kinases and better at phosphorylating transcription factors such as activating-transcription-factor (ATF)2, Elk-1 and SAP-1 (Figure 2) [4].

The  $\alpha$ ,  $\beta$  and  $\delta$  forms are widely expressed, whereas  $\gamma$  is limited to muscle. p38 $\alpha$  is strongly expressed in monocytes, macrophages, CD4<sup>+</sup> T cells, neutrophils and endothelial cells; p38 $\beta$  is only expressed at levels comparable to  $\alpha$  in endothelial cells [5]. Macrophages, CD4<sup>+</sup> T cells and neutrophils all express comparable levels of  $\alpha$  and  $\delta$ . The role of p38 $\gamma$  and  $\delta$  in inflammation is unknown. The anti-inflammatory effects of the pyridinyl imidazole inhibitors result from inhibition of both  $\alpha$  and  $\beta$  forms; the relative importance of  $\beta$  is unknown. A major function of p38 $\alpha$  (and probably  $\beta$ ) is the post-transcriptional regulation of expression of inflammatory response genes.

Inflammatory stimuli activate four major intracellular signalling pathways: the nuclear factor- $\kappa$ B (NF $\kappa$ B) pathway, and the three MAPK pathways 2., 3., 6., 7.. The MAPKs are inactivated by phosphatases, especially the dual specificity phosphatases, also called MAPK phosphatases. Dual specificity phosphatase-1 is induced by inflammatory stimuli and corticosteroids 7., 8.. All four pathways drive transcription of inflammatory genes. The promoter regions of these genes have binding sites for a limited set of transcription factors: sites for NF $\kappa$ B, AP-1, C/EBP $\beta$  and Ets family members are common. JNKs phosphorylate and activate c-jun and ATF2, which are components of AP-1 binding complexes. ERK activates C/EBP $\beta$  and AP-1, and all three MAPKs can phosphorylate Ets transcription factors such as Elk-1 and SAP-1 [6]. The p38 pathway is involved in transcriptional control, but the precise mechanisms are unclear. Several transcription factors can be phosphorylated by p38 (Figure 2) 2., 6., and the downstream kinases might

also be involved. MAPK and stress kinase-activated protein kinase (MSK) 1 and 2 activate the cAMP response element binding protein [9] and the p65 subunit of NF $\kappa$ B [10], and also phosphorylate histone H3 and high mobility group protein 14 [11]. MAPKAPK-2 phosphorylates serum response factor (Figure 2) [6].

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## Section snippets

### Post-transcriptional regulation

The p38 pathway stabilises inflammatory response protein mRNAs 6., 7. and promotes their translation [3'] through AU-rich elements (AREs) in the 3' untranslated region of the mRNAs 12., 13., 14.. These contain repeats of the motif AUUUA, often in a U-rich context, and are present in mRNAs of cytokines, inflammatory proteins and oncogenes 6., 7., 15.. They were discovered as instability elements but, in the case of inflammatory response mRNAs, the instability is countered by signalling in the ...

## Involvement of p38 MAPK in processes other than inflammation

Strong stimulation of p38 MAPK occurs through inflammatory stimuli acting via receptors of the Toll-like and IL-1 receptor family and the TNF receptor family, and as a result of physical cell stress. However, the pathway is ubiquitous, and is activated by growth factors, G-protein coupled receptors, immune receptors and other stimuli 2., 3. Deletion of the p38 $\alpha$  gene (reviewed in [6]) was early-embryonic lethal owing to a defect in placentation. Advanced embryos showed failure of ...

## Inhibitors of p38 *in vivo*

The diaryl imidazoles (e.g. SB203580 and SB202190) not only inhibited inflammatory cytokine production from cells *in vitro* but were also anti-inflammatory in animal models of rheumatoid arthritis [3]. The potential of p38 as a drug target has led to development by pharmaceutical companies of a large collection of inhibitors. Some are based on the original structures (Figure 1), either with different substituents to the imidazole core or by replacing this core with different heterocycles [52]. ...

## Conclusions

The success of anti-TNF $\alpha$  therapy in rheumatoid arthritis, and the dependence of TNF $\alpha$ , IL-1 and COX-2 expression upon p38 activity, have made rheumatoid arthritis a target disease for p38 inhibitors. However, any inflammation strongly dependent upon the two primary cytokines (i.e TNF $\alpha$  and IL-1) will be reduced by p38 blockade. Adverse effects of inhibition are hard to anticipate but immunosuppression is likely. If p38 inhibition has unacceptable toxicity, then MAPKAPK-2 and downstream ...

## References and recommended reading

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